

Tethered 6U CubeSat

Tether range, relative attitude, state of the art and 3D imaging

CUBICS 2026 · Stream 1 · Team meeting, October 2026

One tether, two 3U halves of 4 kg, circular orbit at 550 km.

Four answers for today

Tether range	Vertical only. 500 m to 1 km is feasible: 3.6 to 7.2 mN of tension, about 3.5 h to deploy.
Relative attitude	Exploit the mechanics: the tether and its anchor joint give the direction between the halves, without relying on electronics.
State of the art	Tethers have flown since 1966. Most failures happen during deployment.
3D photos	Not with a 0.5 to 1 km tether baseline: height errors near 1 km. Yes with fore/aft camera pointing: about 3.5 m.
Lifetime	Reel the tether in when not in use: about 6 years instead of 1.6 years at 550 km.

The tether is held taut by the gravity gradient

Circular orbit of radius $a = R_{\oplus} + h$:

$$n = \sqrt{\frac{\mu}{a^3}} = 1.0948 \times 10^{-3} \text{ rad/s}, \quad P = \frac{2\pi}{n} = 95.6 \text{ min.}$$

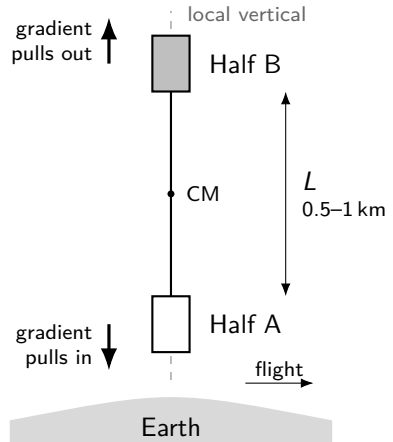
A body forced to turn at rate n at radius $a + l$ lacks

$$n^2(a + l) - \frac{\mu}{(a + l)^2} \simeq 3 n^2 l$$

of inward acceleration. The tether supplies it:

$$T = 3 n^2 m_{\text{eq}} L, \quad m_{\text{eq}} = \frac{m_1 m_2}{m_1 + m_2} = 2 \text{ kg.}$$

$$T = 3.6 \text{ mN (500 m)} \quad T = 7.2 \text{ mN (1 km)}$$



Only the vertical direction separates the two halves

Relative motion in the orbiting frame (x radial, y along track, z out of plane):

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x, \quad \ddot{y} + 2n\dot{x} = f_y, \quad \ddot{z} + n^2z = f_z.$$

Direction	Free motion	Tether
Radial (x)	$+3n^2x$: the halves move apart	taut
Along track (y)	no restoring or separating term	slack
Out of plane (z)	$-n^2z$: the halves move together	slack

A tilt φ from the vertical swings back (constant L):

$$\ddot{\varphi} + 3n^2 \sin \varphi \cos \varphi = 0 \Rightarrow \omega_{\text{lib}} = \sqrt{3} n, \quad 2\pi/\omega_{\text{lib}} = 55.2 \text{ min.}$$

A diagonal tether is a swing, not a configuration. Holding $\varphi = 30^\circ$ at 1 km needs $3n^2 m_{\text{eq}} L^2 \sin \varphi \cos \varphi = 3.1 \text{ N m}$, about 3000 times a CubeSat reaction wheel.

500 m to 1 km with one tether: the key numbers

Quantity	Expression	$L = 500 \text{ m}$	$L = 1 \text{ km}$
Tension	$3n^2 m_{\text{eq}} L$	3.6 mN	7.2 mN
Elastic stretch	$TL/(EA)$	0.6 mm	2.3 mm
Tether mass, $\varnothing 0.2 \text{ mm}$	$\rho_c AL$	22 g	44 g
Tether frontal area	$d L$	0.10 m^2	0.20 m^2
Controlled deployment, $\lambda = n/2$	$\ln(L/L_0)/\lambda$	3.2 h	3.5 h
Controlled retrieval, $\lambda = n/5$	$\ln(L/L_0)/\lambda$	7.9 h	8.8 h
Orbit life at 550 km, deployed	drag model	$\approx 2.5 \text{ yr}$	$\approx 1.6 \text{ yr}$

$EA = 3142 \text{ N}$, $L_0 = 1 \text{ m}$, mean solar activity. Stowed: $\approx 6 \text{ yr}$.

Once deployed, the length stays fixed

The winch sets L . More tension does not lengthen the tether; it only pulls it tighter.

Elastic stretch

$$\Delta L_{\text{el}} = \frac{T L}{EA}$$

$$\leq \mathbf{2.3 \text{ mm}}$$

at 1 km and full tension

Natural swing

$$2\pi/(\sqrt{3} n)$$

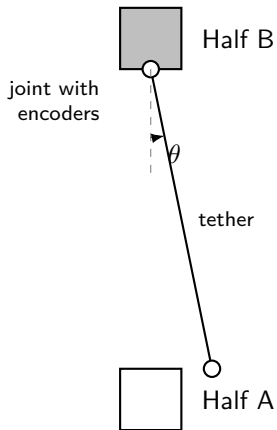
55 min

period around the vertical

The tension is fixed by the length: $T = 3n^2 m_{\text{eq}} L$. Paying out faster only adds swing and braking load.

Relative attitude: exploit the mechanics

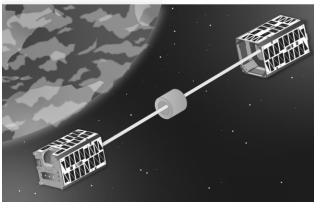
- The tether is a physical link: it already points from one half to the other.
- A passive joint at each anchor, with angle encoders, reads that direction directly.
- The measurement is mechanical: it does not drift and does not depend on extra electronics.
- The joined state gives the mechanical zero for calibration.



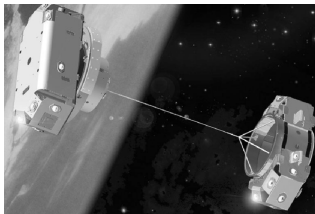
Tether heritage: deployment is the main risk

Mission	Year	Length	Result
Gemini XI	1966	30 m	First tether between two spacecraft
SEDS-1	1993	20 km	Full deployment; end mass deorbited
SEDS-2	1994	19.7 km	Closed-loop braking law validated
TSS-1R	1996	19.7 km	Tether broke by an electrical arc
TiPS	1996	4 km	Libration damped from 40° to 7.5°
ATE _x	1999	22 m of 6.2 km	Tether went slack; deployment stopped
YES2	2007	31.7 km	Longest structure deployed in orbit
TEPCE	2019	1 km	Two CubeSat halves split by a stacer spring at 4 m/s

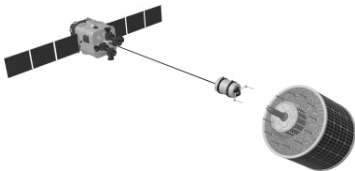
Previous tethered systems



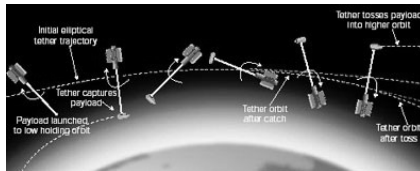
Two CubeSats joined by a tether



Two end bodies joined by a tether



Tether between a spacecraft and a target



Momentum exchange: catch and toss

Where our proposal fits

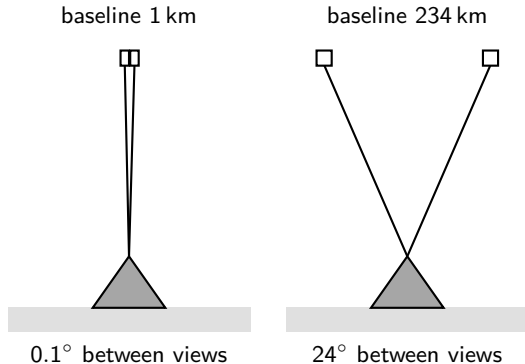
Already demonstrated

- Splitting a CubeSat in two with a spring (TEPCE, 2019)
- Kilometre-class deployment with braking (SEDS-2, TiPS)
- Gravity-gradient stabilisation along the vertical
- Libration damping with the winch

New in this proposal

- One tether with a winch that both deploys and reels in
- Tether direction measured at the anchor
- Tether reeled in between campaigns to extend the mission
- Earth imaging from both halves

3D needs two very different viewing angles



A point of height h shifts between the photos by

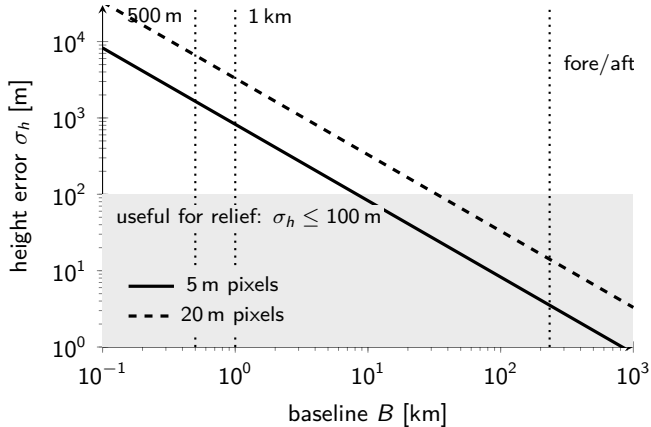
$$p = h \frac{B}{H},$$

so the height error is

$$\sigma_h = \frac{\sigma_p}{B/H}.$$

B baseline, $H = 550$ km, σ_p matching error (0.3 pixel = 1.5 m with 5 m pixels).

A 0.5 to 1 km baseline gives height errors near 1 km



Tether baseline, even if horizontal: $B/H = L/H$.

- 1 km: $\sigma_h \approx$ **825 m**
- 500 m: $\sigma_h \approx$ **1.65 km**
- A 1 km mountain shifts **0.36 pixel**.

A vertical tether gives no stereo at all: both cameras look along the same line.

What works: pointing the cameras

Option	B/H	σ_h (5 m / 20 m pixels)	Notes
Fixed, A $+12^\circ$ / B -12°	0.43	3.5 m / 14 m	Sites under the track
Gimbal $\pm 30^\circ$	0.40	3.8 m / 15 m	Chosen sites, 2–4 days
Tether baseline, 1 km	1.8×10^{-3}	825 m / 3.3 km	Not usable

The 3D comes from the satellite moving 234 km along its orbit in 31 s, not from the tether.

Risks

Retrieval	Reeling in amplifies the swing ($-2\dot{L}/L > 0$). It needs libration control and a slow rate: 7.9 to 8.8 h.
Lifetime	Reel the tether in when not in use: about 6 yr instead of 1.6 yr at 550 km. Deploy it at end of life to reenter faster.
Single tether	One cut by a small debris particle ends the mission. Use a braided or multi-strand line.
Debris and admissibility	A 1 km object has a larger collision cross-section. Confirm with the CSA.
Model accuracy	Orbit lives come from a simple drag model; a formal STELA or DRAMA study is needed.

Decisions for the team

- 1 **Tether length:** 500 m or 1 km, with the tether stowed between campaigns.
- 2 **Relative attitude:** joint with encoders at each anchor as the main measurement.
- 3 **Imaging claim:** drop tether stereo; use fore/aft or gimbal cameras for 3D.
- 4 **Questions to the CSA by 9 November:** launch altitude, tether admissibility, deorbit case.
- 5 **Analysis:** formal orbit-life study with STELA or DRAMA.